

# JETT KAFKA

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Portfolio: <https://jettkafka.github.io/engineering-portfolio/>

## PROFILE

Electrical engineering student at McGill University (eligible for internship via McGill EIP) with a 4.0 CGPA and a constant drive to take on and solve harder problems. Built engineering skills beyond the classroom through 19 independent hardware projects and McGill Rocket Team avionics membership; gained hands-on experience in PCB design, embedded systems, hardware testing, and debugging. Beyond engineering, brings 3 summers of experience as a hockey instructor and camp counsellor, developing the communication, leadership, and judgment needed to mentor and manage group dynamics. Seeking a Summer 2027 internship in embedded systems, hardware design, or firmware engineering.

## EDUCATION

|                                                                                                          |                                       |
|----------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>McGill University</b><br><i>Bachelor of Engineering, Electrical Engineering</i>                       | <b>Montreal, QC</b><br>2025 - Present |
| <ul style="list-style-type: none"><li>Member of the McGill Rocket Team Avionics Radio Division</li></ul> |                                       |

## ENGINEERING EXPERIENCE & PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                       |
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| <b>McGill Rocket Team</b><br><i>Avionics Radio Division</i>                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Montreal, QC</b><br>2025 - Present |
| <ul style="list-style-type: none"><li>Assembled and debugged the SRADIO PCB, a LoRa-based radio board using the SX1262 transceiver for rocket-to-ground communication.</li><li>Interpreted component datasheets to ensure correct PCB assembly and operating conditions, preventing component damage and hardware faults during bring-up.</li><li>Collaborated in a two-person project team, contributing to technical decisions, troubleshooting, and project execution.</li></ul> |                                       |

### 4-Bit Computer (Independent Project)

- Built a breadboard-based 4-bit computer integrating a program counter, EEPROM program ROM, control logic, RAM, registers, and ALU.
- Debugged timing and synchronization across digital ICs while implementing low-level computer architecture and control logic.

### 2D Servo LiDAR Mapper (Independent Project)

- Developed a 2D scanning system using a TF Mini Plus LiDAR sensor, closed-loop servos, and Arduino-based control to map distance measurements across a field of view.
- Progressed the system from breadboard prototype through PCB implementation, testing, and final integration.
- Designed and 3D printed a custom enclosure and mounting hardware in Fusion 360.

### Successive Approximation Register ADC (Independent Project)

- Built and tested a successive-approximation ADC using digital control logic, an R/2R ladder DAC, comparator feedback, and clocked registers.
- Implemented manual and continuous conversion modes to evaluate ADC behavior under varying analog inputs.

## ADDITIONAL EXPERIENCE

|                                                                                                                                                                                                                                                                                                               |                                          |
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| <b>Solly's Craft Soda</b><br><i>Brand Ambassador</i>                                                                                                                                                                                                                                                          | <b>Toronto, ON</b><br>July - August 2026 |
| <ul style="list-style-type: none"><li>Adapted communication and approach based on customer reactions to improve engagement and product interest.</li><li>Interacted with 4000+ potential customers.</li><li>Promoted products through in-store sampling, customer engagement and product education.</li></ul> |                                          |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Camp Manitou</b><br><i>Ball Hockey Instructor and Cabin Counsellor</i>                                                                                                                                                                                                                                    | <b>Parry Sound, ON</b><br>Part Time June - August 2024 - 2026 |
| <ul style="list-style-type: none"><li>Learned coordination, communication and conflict resolution skills through interacting with 12-13 kids and ensuring they've had the best experience of their life</li><li>Created engaging and accommodating games to campers of all ages and hockey levels.</li></ul> |                                                               |

|                          |                    |
|--------------------------|--------------------|
| <b>Bill Bolton Arena</b> | <b>Toronto, ON</b> |
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- Enforced game rules and safety policies while resolving on-ice disputes and promoting sportsmanship among youth players.

## TECHNICAL SKILLS

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**Programming:** C++, Python, Assembly

**Embedded Systems:** SPI, I<sup>2</sup>C, UART, Timers, EEPROM, Flash

**PCB Design:** Altium Designer, EasyEDA

**Hardware & Test Equipment:** Soldering, Oscilloscopes, Spectrum Analyzers, Multimeters

**CAD:** Fusion 360 (3D CAD)

**Development Tools:** Git, Visual Studio Code, Cursor

**Productivity:** Microsoft Excel, Microsoft Word